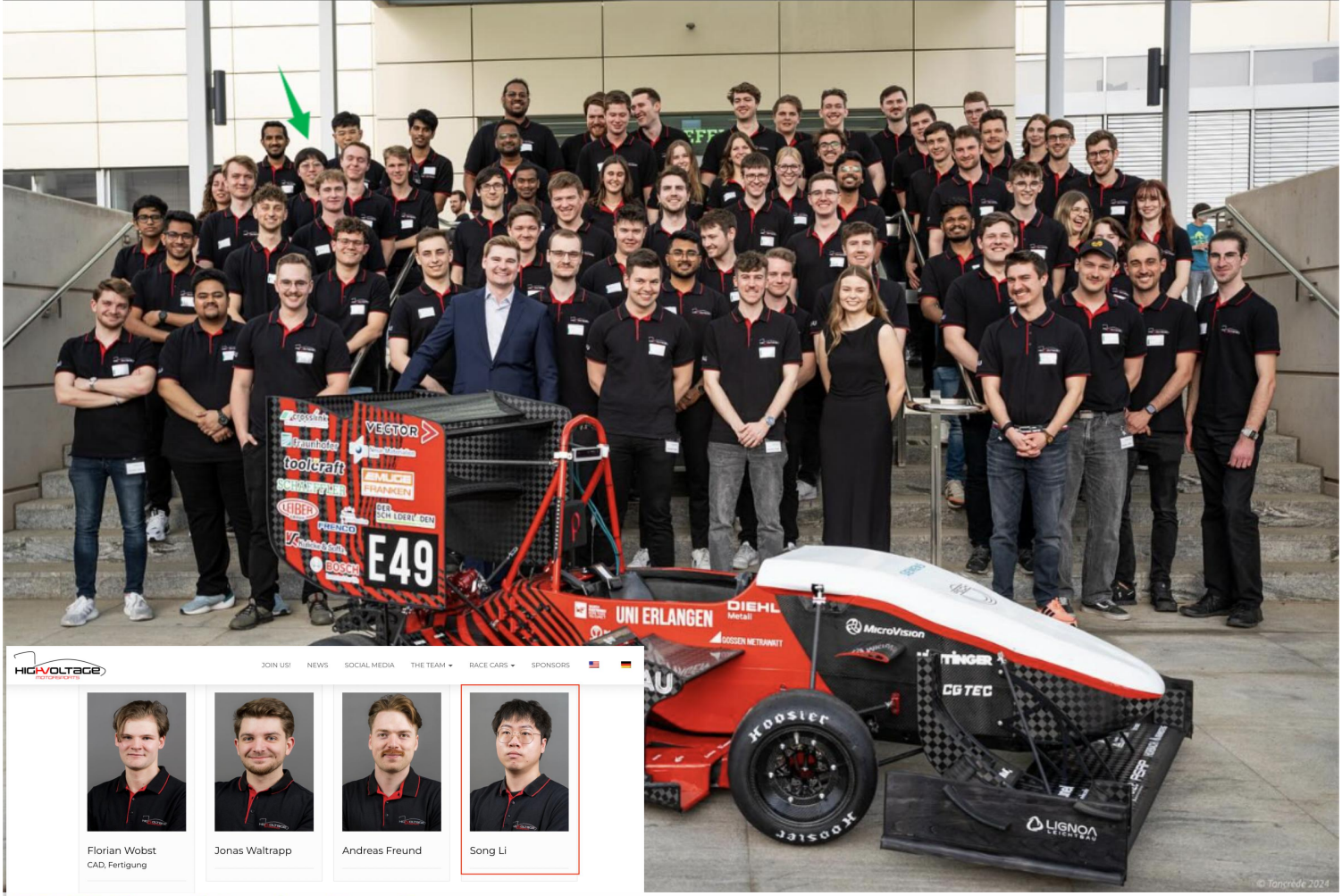


FSD23 Overall Scoring Results

FSD23 Overall Scoring Results [ver. 2 - 2023-08-22 21:50]

Revised overall results, due to updated results for Trackdrive.

Search:

Car	City / University	Design	DV ACC	DV SkidPad	AutoX	TRD
95	Göteborg Chalmers	90,00	0,00	72,17	0,00	200,00
110	Augsburg UAS	100,00	0,00	0,00	0,00	113,18
14	Budapest TU	100,00	0,00	16,21	0,00	96,69
50	Lisboa IST	100,00	9,64	41,09	43,56	85,16
322	Napoli UNINA	95,00	0,00	3,50	10,00	40,00
120	Karlsruhe KIT	150,00	64,05	75,00	0,00	15,00
42	Ulm UAS	80,00	0,00	0,00	0,00	5,00
31	München TU	100,00	75,00	0,00	0,00	5,00
97	Schweinfurt UAS	55,00	0,00	0,00	0,00	0,00
172	Dortmund TU	60,00	0,00	0,00	0,00	0,00
44	Deggendorf IT	55,00	0,00	3,50	10,00	0,00
62	Regensburg OTH	70,00	0,00	0,00	0,00	0,00
49	Erlangen U	70,00	0,00	0,00	0,00	0,00
101	Mannheim DHBW	80,00	0,00	0,00	0,00	0,00

DC24 Overall Scoring Results

DC24 Overall Scoring Results [ver. 1 - 2024-08-18 23:51]

Protest time end 2024-08-20 22:00 CEST

Search:

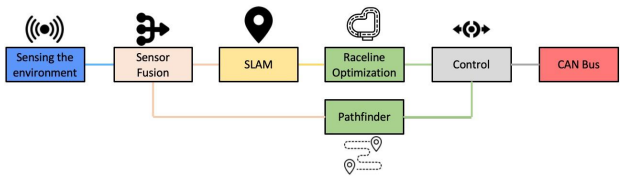
Car	City / University	Design	DV ACC	DV SkidPad	AutoX	TRD
33	Zürich ETH	115,00	14,16	71,43	99,62	200,00
95	Göteborg Chalmers	90,00	55,12	59,44	93,68	199,03
90	Milano PT	110,00	0,00	60,75	74,67	181,49
66	Melbourne Monash	120,00	8,92	3,75	10,00	76,55
54	Barcelona ETSEIB	100,00	3,75	0,00	0,00	51,94
240	Darmstadt TU	65,00	3,75	3,75	10,00	45,00
49	Erlangen U	85,00	0,00	0,00	0,00	20,00
59	Dresden TU	100,00	75,00	75,00	100,00	15,00
42	Prague CTU	100,00	43,28	67,69	73,21	10,00
22	Leuven KU	95,00	0,00	0,00	10,00	5,00
24	Karlsruhe KIT	150,00	73,16	69,98	56,18	5,00
313	Berlin TU	65,00	0,00	0,00	0,00	0,00

- The group photo is from the rollout of the 2024 new car at the Schaeffler Campus (https://www.instagram.com/p/C6g8LSSLzVn/).
- The individual portrait in the bottom-left corner of the group photo can be found at: https://voltages.de/en/team/team-electric/2025-sigma/.
- The two result tables compare the performance improvement during my participating season at FSG 2024 compared to the previous year (https://www.formulastudent.de/fsg/results/2024).
- The picture on the right shows the LiDAR SLAM testing in the open Norma supermarket parking lot near Wetterkreuz.

HIGHVOLTAGE MOTORSPORTS

Friedrich-Alexander-Universität Erlangen-Nürnberg
Faculty of Engineering

Software Pipeline



Environmental Sensing
- Real-time data acquisition from LiDAR, cameras, and IMU provides a comprehensive view of the surroundings.

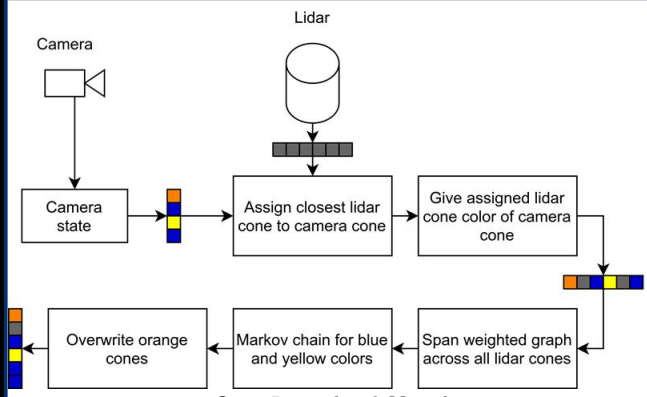
Sensor Fusion
- Integrating data from LiDAR, cameras, and IMU enhances environmental awareness and improves accuracy.

SLAM (Simultaneous Localization and Mapping)
- Creates detailed track maps and accurately localizes the vehicle during navigation.

Raceline Optimization
- Employs loop closure detection to minimize errors and optimize the racing line for speed and precision.

Lateral Control
- Ensures stable and responsive vehicle movement through accurate steering control.

Camera-LiDAR Fusion



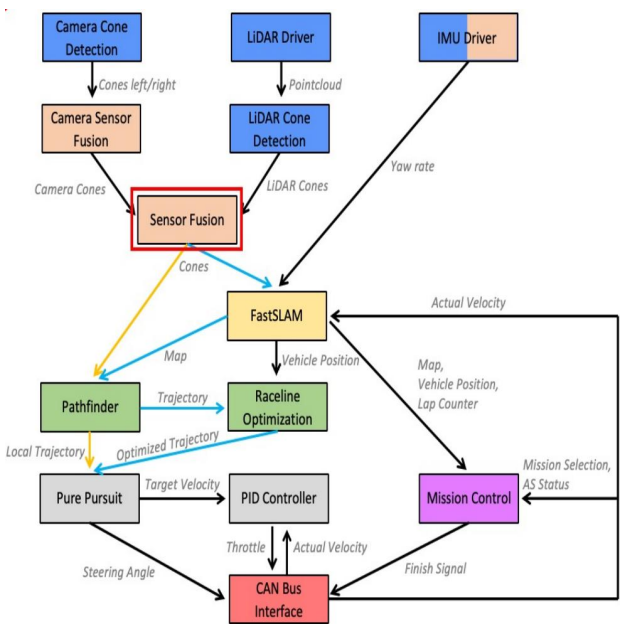
Cone Detection & Mapping

- Camera: Powered by YOLO object detection, precisely identifies cone locations.

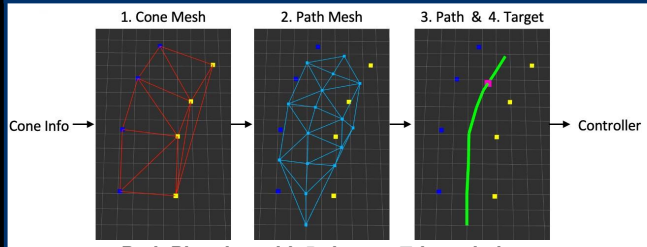
- LiDAR: Provides accurate distance measurements.

- Sensor Integration: Combines camera and LiDAR strengths for reliable cone detection and mapping, enabling efficient track navigation.

Sensor Fusion and Control System Diagram



Pathfinder



Path Planning with Delaunay Triangulation

- Method: Uses Delaunay triangulation to generate efficient routes by prioritizing short edges and large angles.

- Benefits: Ensures smooth, collision-free navigation and optimizes paths for dynamic track conditions.

Cones Data Augmentation